

Learning Z

The partition function — and the discovery that you already own almost every book

A re-audit across **Pure_Maths_TC_SelfStudy**, **TensoQi101**, **QBIT-HIS** and **qm-qi-qml** — 681 books, every claim verified by opening the file.

$$Z = \text{Tr}(T^N)$$

$$\begin{pmatrix} e^{\beta(J+h)} & e^{-\beta J} \\ e^{-\beta J} & e^{\beta(J-h)} \end{pmatrix}$$



$$\lambda_+^N + \lambda_-^N$$

2^N configurations
collapse to two numbers

Pure_Maths

153 books

measure, analysis,
Feynman's stat mech

TensoQi101

121 books

QM, quantum info,
DFT, Shankar

QBIT-HIS

66 books

quantum stat mech,
stat mech of the cell

qm-qi-qml

341 books

QML, MIT 8.04
(little new for Z)

What version 1.0 got wrong. v1.0 reported that Modules 5–8 of **Pure_Maths_TC_SelfStudy** were empty, and sent you shopping for five books.

Four of those five were already on your shelf, dated **March 2024** — eighteen months before the audit. **Feynman's Statistical Mechanics** was sitting in **Module_8**. The corrected finding is in §1, and the gap list has collapsed from five books to two.

The standout find. **Shankar §21.2** (*Principles of QM*, p. 627, in **TensoQi101**) solves the 1D Ising partition function *and* maps it onto the quantum path integral. **Feynman ch. 1, 3 and 5** does the same arc from the other direction.

You own two independent treatments of the exact calculation in §5 of this dossier — one from quantum mechanics, one from statistical mechanics.

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This version supersedes v1.0

§1 is the correction — read it first if you saw v1.0. §2 is the corrected four-library audit. §4 is the revised route. §7 shows you already own two treatments of the §5 calculation. §8 scopes the QBIT ambition honestly. §9 is the shopping list, now two books.

§3, §5 and §6 are unchanged from v1.0 — they were independently computed and remain correct.

1

Correction to version 1.0

Version 1.0 of this dossier made a claim that was false, and acted on it. The correction comes first because it changes what you should do.

What v1.0 said, and what is actually true	
v1.0 claimed	Verified today
“Modules 3–8 are empty directories.”	Modules 3 and 4 are empty. Modules 5, 6, 7 and 8 are not , and were not in 2024.
“There is no statistical mechanics on this shelf.”	Feynman, <i>Statistical Mechanics: A Set of Lectures</i> (371 pp) is in Module_8.
“No probability theory, no stochastic processes.”	Le Gall, <i>Measure Theory, Probability and Stochastic Processes</i> (409 pp) is in Module_8; Lauritzen is beside it.
“Measure theory only via Kuttler ch. 20–24.”	Axler, <i>Measure, Integration & Real Analysis</i> (430 pp) is in Module_5 — by the same author as your linear algebra text.
“Buy five books.”	Four of the five were already yours. File dates: March 2024.

IN PLAIN WORDS

The lesson is the same one Part VIII made about the sequencing run, turned on its author: a method can be correctly executed and still produce a confidently wrong answer, because the failure was in the *model of where errors come from*, not in the arithmetic. I searched for titles containing physics keywords and concluded absence from silence. Feynman’s book is filed under *Mathematical Statistics*; Axler’s measure theory is filed under *Real Analysis*. Neither name contains the word I was grepping for.

Everything in v1.0 §§2–8 — the three faces of Z, the Ising calculation, the Monte Carlo failure, the bridge to Part IX — was independently computed and remains correct. What follows replaces the audit, the route and the shopping list.

2 The four-library audit

Library	books	size	what it contributes to Z
Pure_Maths_TC_SelfStudy	153	552 MB	operators (Axler LADR), measure (Axler MIRA, Kuttler, Le Gall), Feynman's statistical mechanics
TensoQi101	121	462 MB	the entire quantum side: Shankar , Sakurai, Nielsen & Chuang, Schumacher, 4 DFT texts
QBIT-HIS	66	1.1 GB	quantum statistical mechanics (Attard, Bru & Pedra) and statistical mechanics of the cell (PBoC2, Bressloff)
qm-qi-qml	341	404 MB	quantum machine learning and MIT 8.04 course material. Little that is new for Z
total	681	2.5 GB	

2.1 Coverage map, corrected

what Z needs	status	which book, and which library
operators, spectral thm, trace	FULL	Axler <i>LADR</i> §7.B, 7.C, 10.A [PM]
measure & Lebesgue integral	FULL	Axler <i>MIRA</i> 2022 · Le Gall · Kuttler 20–24 [PM]
probability & stochastic proc.	FULL	Le Gall · Lauritzen [PM] · Bressloff 1–2 [QB]
complex analysis, saddle point	FULL	Radozycki Part III [PM]
classical statistical mechanics	FULL	Feynman ch. 1, 5 [PM] · PBoC2 ch. 6 [QB] · Shankar §21.2 [TQ]
quantum mechanics	FULL	Shankar · Sakurai · Griffiths · Binney & Skinner [TQ]
density operators, vN entropy	FULL	N&C §2.4, §11.3 · Schumacher ch. 8, §19.5 [TQ]
quantum statistical mechanics	FULL	Attard 2016 · Feynman ch. 2–3 · Bru & Pedra 2020 [QB]
electronic structure, $U(x)$	FULL	4 DFT texts incl. a Harvard practical guide [TQ]
open quantum systems	PARTIAL	N&C ch. 8 (Kraus, not HEOM) · Khrennikov (<i>contested</i>)
Markov-chain mixing times	NONE	— nothing in any of the four libraries
molecular simulation, free energy	NONE	— GROMACS 2023.3 is built, but no text

[PM] Pure_Maths [TQ] TensoQi101 [QB] QBIT-HIS

Figure 1. Corrected coverage. Nine of twelve layers are fully covered by books you already own. The line separates what you can study tomorrow from what must be acquired. Compare v1.0, which showed three of ten red.

2.2 The three finds

1 · QBIT-HIS — you had already built the library for this

66 books, 1.1 GB, and the most relevant folder you own. For Z specifically:

Book	pp	role
Attard, <i>Quantum Statistical Mechanics from First Principles</i> (2016)	285	exactly the title
Attard, <i>Entropy Beyond the Second Law</i> (2023)	—	non-equilibrium; Part IX Gap 2
Bru & de Siqueira Pedra, <i>C*-Algebras and Math. Foundations of Quantum Stat. Mech.</i> (2020)	497	the rigorous destination
Phillips et al., <i>Physical Biology of the Cell</i> 2ed (2012)	1089	ch. 6 <i>Entropy Rules!</i> — Boltzmann distribution, free-energy minimisation, statistical mechanics of gene expression
Bressloff, <i>Stochastic Processes in Cell Biology</i> Vols 1–2 (2022)	773 +	the definitive treatment
Barato & Seifert, thermodynamic uncertainty relation (1502.05944)	11	cited in Part IX — you had the preprint
Khrennikov, <i>Open Quantum Systems in Biology</i> . . . (2023)	371	Part IX Tier 3 (<i>idiosyncratic; read as a proposal</i>)
Protein-DNA-Search/ — Mirny 2009, first-passage under restart	7 files	statistical mechanics of molecular search

2 · TensoQi101 — the quantum half, complete

I grepped every PDF over 1 MB for the phrase “*partition function*”:

hits	book
14	Shankar, <i>Principles of Quantum Mechanics</i> 2ed (1994) — 696 pp
8	Binney & Skinner, <i>The Physics of Quantum Mechanics</i> (Oxford 2013)
6	Schumacher & Westmoreland (2010) · Sakurai & Napolitano 3ed (2021)
2	Nielsen & Chuang (710 pp) · Vedral

Plus the operator-theoretic apparatus: **N&C §2.1.8** (operator functions — how $e^{-\beta\hat{H}}$ is defined), **§2.4** (density operator), **§11.3** (von Neumann entropy), **ch. 8** (quantum noise: Kraus/operator-sum, amplitude and phase damping — 101 occurrences of Lindblad/master-equation/Kraus). Schumacher **§19.5** is titled *Entropy and thermodynamics*.

3 · qm-qi-qml — large, but not for this

341 files, of which roughly ten are books; the remainder is MIT 8.04 OCW material (Zwiebach). A keyword search for statistical mechanics, thermodynamics, entropy, measure, probability, stochastic, Ising, Monte Carlo, Boltzmann, Gibbs, partition, open quantum systems, density matrix and Lindblad returned **no title matches**. Valuable for quantum machine learning; not a source for Z.

3 What Z is: three faces

$$Z = \int e^{-\beta U(x)} dx \quad (\text{classical}) \qquad Z = \text{Tr}(e^{-\beta \hat{H}}) = \sum_n e^{-\beta E_n} \quad (\text{quantum})$$

These look like different objects. They are one object wearing three hats, and each hat corresponds to a different part of your shelf.

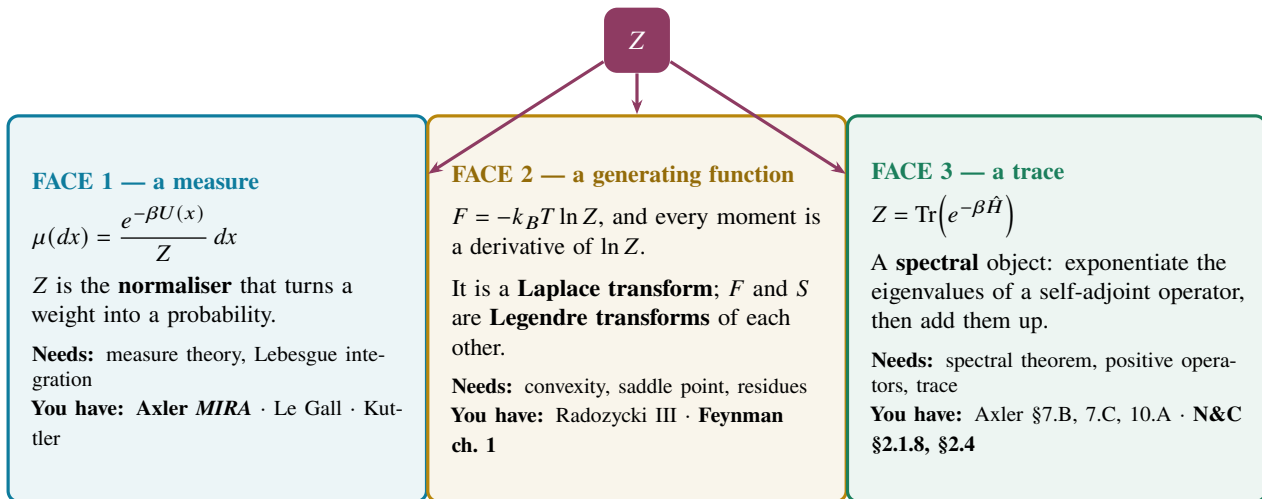


Figure 2. The three faces, each mapped to a book you already own. **Start with Face 3** — it is the one your shelf covers most completely, and it is the one that turns into a concrete calculation fastest.

Why Face 3 first, precisely

$e^{-\beta \hat{H}}$ is not a formal symbol. It is an operator, and it exists because:

1. $\hat{H}$ is **self-adjoint**, so by the **spectral theorem** (Axler §7.B, p. 217 for the complex case) it has an orthonormal eigenbasis with real eigenvalues E_n ;
2. you may therefore *define* $e^{-\beta \hat{H}}$ by exponentiating each eigenvalue;
3. the result is a **positive operator** (Axler §7.C, p. 225) — which is exactly why $Z > 0$ and why μ is a genuine probability measure;
4. and Tr is basis-independent (Axler §10.A, p. 296–299), so $\text{Tr}(e^{-\beta \hat{H}}) = \sum_n e^{-\beta E_n}$ without choosing coordinates.

Four pages of Axler and you have constructed the quantum partition function. No physics book required for this step — although Nielsen & Chuang §2.1.8 (“Operator functions”) states the same construction in one page, and Feynman ch. 2 then uses it to define the density matrix of a thermal state.

IN PLAIN WORDS

The partition function is one number that summarises every way a system can arrange itself, weighted by how likely each arrangement is. Once you have it, every measurable quantity — energy, entropy, heat capacity, binding affinity — is a derivative of its logarithm. That is why it is called the *partition* function: it tells you how probability is partitioned across states.

4 The route, corrected — eight stages, all but one from your own shelves

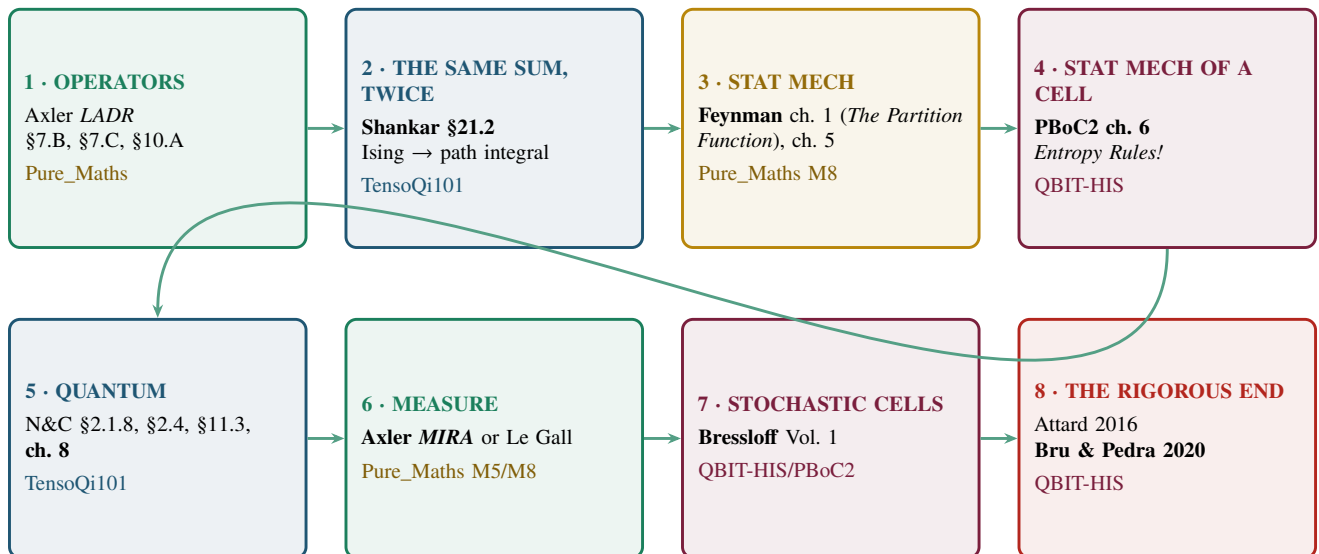


Figure 3. The corrected route. Every stage is a book you already own. Only the Markov-mixing material (§9) has to be acquired, and it is free.

4.1 Stage 1 in detail — the four pages that matter

Axler <i>LADR</i> section	p.	Why $Z = \text{Tr}(e^{-\beta\hat{H}})$ needs it
7.B The Spectral Theorem	217 / 219	lets you <i>define</i> $e^{-\beta\hat{H}}$ at all — you exponentiate eigenvalues, which needs an orthonormal eigenbasis to exist
7.C Positive Operators	225	$e^{-\beta\hat{H}}$ is positive. Hence $Z > 0$, and $e^{-\beta U}/Z$ is a probability density
10.A Trace	296–299	basis-independence makes $\text{Tr}(e^{-\beta\hat{H}}) = \sum_n e^{-\beta E_n}$ a definition, not a coincidence

Cross-check as you go

Read **Nielsen & Chuang §2.1.8** immediately afterwards. It states the same construction (“operator functions”) in about a page, in physicists’ notation. Having the same theorem twice, in two vocabularies, is the fastest way to know you have actually understood it rather than memorised one presentation.

5 The worked calculation: 1D Ising, three ways

Everything in this section and the next was computed for this dossier and is reproducible from the companion notebook. It uses numpy only.

5.1 The model, and where the linear algebra enters

N spins $s_i = \pm 1$ on a ring, $E(s) = -J \sum s_i s_{i+1} - h \sum s_i$. Write the Boltzmann weight so each *bond* contributes one factor:

$$e^{-\beta E} = \prod_{i=1}^N \underbrace{\exp\left[\beta J s_i s_{i+1} + \frac{\beta h}{2}(s_i + s_{i+1})\right]}_{T(s_i, s_{i+1})} \quad T = \begin{pmatrix} e^{\beta(J+h)} & e^{-\beta J} \\ e^{-\beta J} & e^{\beta(J-h)} \end{pmatrix}$$

Chaining the factors and closing the ring turns the sum over 2^N configurations into a trace:

$$Z = \sum_{s_1} \cdots \sum_{s_N} T(s_1, s_2) \cdots T(s_N, s_1) = \text{Tr}(T^N) = \lambda_+^N + \lambda_-^N$$

$$\lambda_{\pm} = e^{\beta J} \cosh \beta h \pm \sqrt{e^{2\beta J} \sinh^2 \beta h + e^{-2\beta J}}$$

Notice exactly which theorems did the work

T is symmetric and has positive entries, so **Axler §7.B** gives an orthonormal eigenbasis; **§7.C** says T is a positive operator, so $\lambda_{\pm} > 0$; **§10.A** says the trace is basis-independent, so $\text{Tr}(T^N) = \lambda_+^N + \lambda_-^N$ without choosing coordinates.

A sum over 2^N states collapsed to two numbers, using three sections of a linear algebra textbook you already own.

5.2 Method A versus Method B: they agree to machine precision

N	2^N states	β	Z from $\text{Tr}(T^N)$	Z from brute force
4	16	1.0	1.2123293134×10^2	1.2123293134×10^2
8	256	1.0	9.1604387226×10^3	9.1604387226×10^3
12	4,096	1.0	7.7491438244×10^5	7.7491438244×10^5
16	65,536	0.2	9.0061138775×10^4	9.0061138775×10^4
16	65,536	0.5	4.4784000941×10^5	4.4784000941×10^5
16	65,536	1.0	6.8584536591×10^7	6.8584536591×10^7

Table 1. Worst relative disagreement across all twelve test cases: 7.56×10^{-13} — machine precision. A sum over 65,536 configurations and the trace of a 2×2 matrix are the same number.

5.3 Thermodynamics: everything is a derivative of $\ln Z$

As $N \rightarrow \infty$ only the larger eigenvalue survives, because $(\lambda_-/\lambda_+)^N \rightarrow 0$. At $h = 0$, $\lambda_+ = 2 \cosh \beta J$, so

$$f = -\frac{1}{\beta} \ln(2 \cosh \beta J), \quad u = -J \tanh \beta J, \quad \frac{C}{k_B} = \left(\frac{\beta J}{\cosh \beta J} \right)^2$$

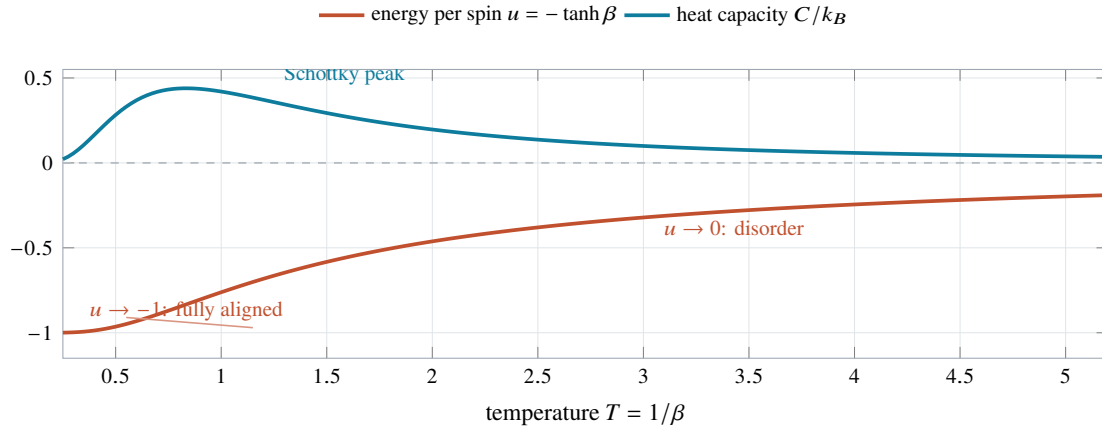


Figure 4. Exact thermodynamics of the 1D Ising chain, from $\ln \lambda_+$ alone. Note there is *no* phase transition: f is smooth for all $T > 0$. That is a theorem — a one-dimensional system with short-range interactions cannot order at finite temperature. The bump in C is a Schottky anomaly, not a critical point.

5.4 The eigenvalue gap is the correlation length

This is the deepest line in the whole calculation, and it is one line of linear algebra:

$$\langle s_0 s_r \rangle = \left(\frac{\lambda_-}{\lambda_+} \right)^r = e^{-r/\xi}, \quad \xi = \frac{1}{\ln(\lambda_+/\lambda_-)}$$

β	λ_+	λ_-	λ_+/λ_-	ξ
0.2	2.04013	0.40267	5.0665	0.616
0.5	2.25525	1.04219	2.1640	1.295
1.0	3.08616	2.35040	1.3130	3.672
1.5	4.70482	4.25856	1.1048	10.034
2.0	7.52439	7.25372	1.0373	27.296
2.5	12.26458	12.10041	1.0136	74.205
3.0	20.13532	20.03575	1.0050	201.714

Table 2. As temperature falls the spectral gap closes and the correlation length grows *exponentially* — from 0.6 to 202 lattice spacings. **Keep this sentence: the spectral gap of the transfer matrix controls how far correlations reach.** It returns in §6 as the reason Monte Carlo fails, and in Part IX §6 as the reason the transfer *operator* is the right object for molecular kinetics.

6 Where Monte Carlo breaks — measured

Now the method everyone actually uses on systems that are *not* exactly solvable: Metropolis. Propose a spin flip, accept with probability $\min(1, e^{-\beta\Delta E})$, run long enough.

That last phrase is where everything goes wrong, and the Ising model lets us measure exactly how wrong.

A perfect test, because we know the answer exactly

At $h = 0$ the model is exactly symmetric under flipping every spin. Therefore

$$\langle M \rangle = 0 \quad \text{exactly, for every } \beta, \text{ every } N.$$

No approximation, no limit. So: run several *independent* chains and see whether they all return zero. **Every chain below runs the identical, correct algorithm. None of them is buggy.**

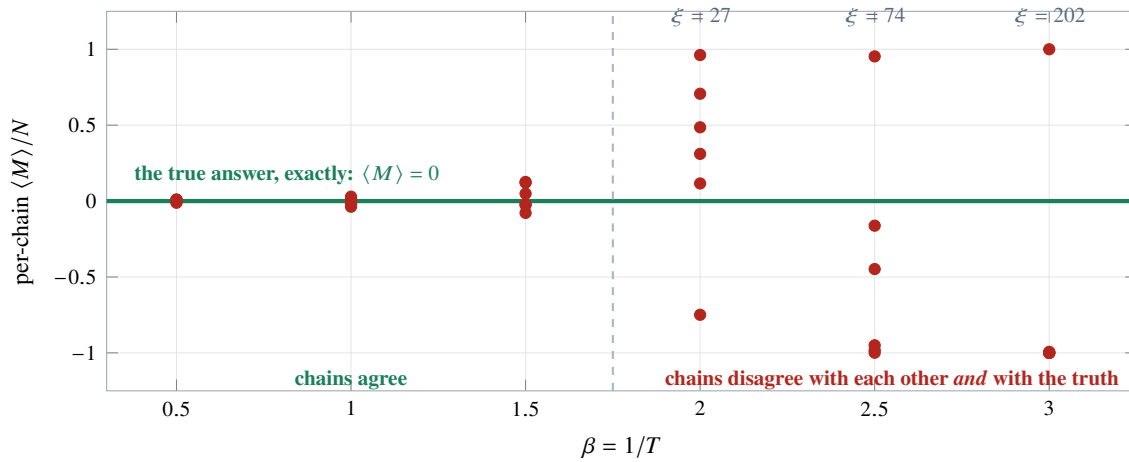


Figure 5. Six independent Metropolis chains, $N = 64$, 6,000 sweeps each. Measured for this dossier. At $\beta = 3$ the chains report ± 1 when the exact answer is 0.

β	ξ	spread across chains	verdict
0.5	1.30	0.021	ok
1.0	3.67	0.066	ok
1.5	10.03	0.203	poor
2.0	27.30	1.710	BROKEN
2.5	74.21	1.952	BROKEN
3.0	201.71	2.000	BROKEN

Table 3. A spread of 2.000 means some chains returned +1 and others -1 — the maximum possible disagreement. An independent 8-chain run at 30,000 sweeps reproduced the same pattern.

Why — and it is the same mechanism as the whole of Part IX

As β grows, ξ grows exponentially (Table 2). Once $\xi \gtrsim N$ the entire ring is a single domain, and to get from “mostly up” to “mostly down” the chain must pass through a high-energy configuration. The waiting time for that is $\sim e^{\beta\Delta E}$ — **exponential in the barrier height**.



This is the quantitative form of the sentence in Part IX §2: “*Monte Carlo converges at $O(n^{-1/2})$ provided it mixes. On a rugged landscape with barriers $\gg k_B T$, it does not mix.*” And it is the argument for Boltzmann generators: rather than *crossing* the barrier, learn a change of variables in which there is no barrier to cross.

**IN
PLAIN
WORDS**

The lesson is not “Monte Carlo is bad”. It is that a method can be provably correct, correctly implemented, and still give you a confidently wrong answer — because “run it long enough” hid an exponential. You would never know, in a system where the true answer was not available. That is the situation in every protein simulation ever run.

7 You own two treatments of this exact calculation

The calculation in §5 is not something I invented for this dossier. Two books on your shelves do it, from opposite directions, and reading both is worth more than reading either twice.

SHANKAR §21.2

Principles of Quantum Mechanics 2ed, ch. 21 *Path Integrals: Part II*, p. 627 — [TensoQi101/Main_Ref_Texts](#)

Arrives from **quantum mechanics**. Verbatim from the text:

eq. 21.2.44 — “Consider the partition function for a quantum system”

“The role of the action in the Feynman integral is played by the energy in the partition function.”

“The role of $\hbar$ is played by T .”

“The dimensionality goes up by 1 as we go from the quantum to the classical problem.”

eq. 21.2.59 — “consider the Ising model in one dimension... The partition function is...”

Then solves for the free energy and correlation function, and maps it back to a quantum problem.

FEYNMAN, *STATISTICAL MECHANICS*

371 pp — [Pure_Maths/Module_8](#)

Arrives from **statistical mechanics**. Chapter structure:

1 Introduction to Statistical Mechanics

§1.1 The Partition Function

2 Density Matrices

§ Density Matrix in Statistical Mechanics (p. 47)

3 Path Integrals

§3.1 Path Integral Formation of the Density Matrix

(p. 72)

§3.4 Variational Principle for the Path Integral

5 Order–Disorder Theory

§5.2 Order–Disorder in One Dimension (p. 130)

§5.3 Approximate Methods for Two Dimensions

§5.4 The Onsager Problem (p. 136)

same

object

Figure 6. Two routes to one calculation. Shankar starts from $\text{Tr}(e^{-\beta\hat{H}})$ and arrives at a classical spin chain; Feynman starts from the classical partition function and arrives at the density matrix as a path integral. **They meet in the middle, at the 1D Ising model.**

What §5 of this dossier adds, and what it does not

The notebook gives you something neither book does: **the calculation done three ways with the numbers agreeing to 7.6×10^{-13} , and then the third method visibly failing.** That is the part you cannot get from reading.

What §5 does not teach, and where to get it:

Not shown in 1D	Where you already have it
Why computing Z is #P-hard — in 1D it is not	Feynman §5.3–5.4: two dimensions, then the Onsager problem
Phase transitions — impossible in 1D with short range	Feynman ch. 5; PBoC2 ch. 6 for the biological versions
Continuous configuration space	PBoC2 on ligand binding; Bressloff Vol. 1
The quantum–classical correspondence	Shankar §21.2 · Feynman ch. 3

8 On QBIT — scoping the ambition honestly

You built TensoQi101 and QBIT-HIS to reach *Quantum Biological Information Theory* and attack three problems with it: mutation, docking, structure prediction. The libraries are excellent. The scoping deserves to be precise, because getting it wrong costs years.

Problem	quantum?	What it actually needs
Protein structure prediction	×	Classical statistical mechanics and geometry. There is no quantum mechanics in AlphaFold. Your route here is PBoC2 → Bressloff → the transfer-operator material of Part IX.
Protein–ligand docking	~	Classical statistical mechanics for the free energy; quantum only to <i>parameterise</i> $U(x)$. That is precisely what your four DFT texts are for — and why GROMACS is on your machine.
Mutation	✓	Genuinely quantum, in one specific place: proton tunnelling in base pairs (Löwdin 1963; Slocombe, Sacchi & Al-Khalili 2022). But most spontaneous mutation is classical — replication error, deamination, oxidative damage.

The honest shape of QBIT

Quantum information theory is the right language for one third of one of your three problems.

That is not a demotion of the ambition — it is a sharpening of it. Two consequences:

- 1. The information-theoretic angle that generalises across all three is Shannon and Kolmogorov, not qubits.** Entropy, mutual information, large deviations and rate functions apply to gene expression, ligand binding and conformational ensembles alike. Nielsen & Chuang **ch. 11** gives you the classical half before the quantum half.
- 2. Where quantum genuinely enters — the tunnelling substrate of mutation — your real obstacle is Part IX's Tier 3:** non-Markovian open-system dynamics at strong coupling, in a structured bath. **N&C ch. 8** is the entry; **Bru & de Siqueira Pedra (2020)** is the rigorous destination; HEOM is in neither and would have to be acquired.

Two books in QBIT-HIS to read with your guard up

Khrennikov, *Open Quantum Systems in Biology, Cognitive and Social Sciences* (2023). The open-systems mathematics is standard and useful. The programme of applying quantum formalism to cognition and social science is **idiosyncratic and not accepted**. Take the formalism; leave the interpretation.

***Towards Quantum Integrated Information Theory* (1806.01421).** Speculative, and built on IIT, which is itself contested. Not a foundation to build on.

By contrast **Marais et al., *The future of quantum biology* (2018)** — also in QBIT-HIS — is a legitimate review and the right place to calibrate what is established versus what is hoped for. Read it before the other two.

IN PLAIN WORDS

The most valuable thing in your QBIT folder is not the quantum material. It is *Physical Biology of the Cell* and Bressloff — statistical mechanics applied to real cells, with numbers. Those two books connect to all three of your target problems. The quantum books connect to one corner of one of them, and that corner happens to be the hardest open problem in the field.

9 What is actually still missing

The v1.0 list had five books. Four were already yours. **Two remain.**

#	Gap	Get	Why
1	Markov-chain mixing times	Levin & Peres , <i>Markov Chains and Mixing Times</i> (the authors host a free PDF)	The only thing that explains §6 <i>quantitatively</i> . You now have Le Gall and Bressloff for stochastic processes, but neither treats mixing time — which is the exact quantity that makes sampling Z fail.
2	Free-energy methods in simulation	Frenkel & Smit , <i>Understanding Molecular Simulation</i>	You have GROMACS 2023.3 compiled and no text for the mathematics it implements. Umbrella sampling, thermodynamic integration, free-energy perturbation — the practical route from Z to a binding affinity.

Optional, only if you pursue the Tier 3 problem

—	HEOM / strong-coupling open systems	Breuer & Petruccione, <i>The Theory of Open Quantum Systems</i>	Goes beyond N&C ch. 8's Kraus formalism into master equations proper.
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Books v1.0 told you to buy that you already own

v1.0 said “buy”	Where it already was
Chandler or Kardar, statistical mechanics	Feynman , <i>Statistical Mechanics</i> — Module_8. Arguably better than either for your purposes, because ch. 2–3 go straight to the density matrix
Grimmett & Stirzaker, probability	Le Gall (409 pp) and Lauritzen — Module_8
Reed & Simon, functional analysis	Axler MIRA (430 pp) — Module_5; and Bru & Pedra for the operator-algebra end — QBIT-HIS
Hall or Sakurai, quantum mechanics	Sakurai & Napolitano 3ed (+ 2 solution manuals), Shankar, Griffiths, Townsend, Binney & Skinner — TensoQi101

10 A twelve-week plan, revised

Weeks	Read	Do	Question to answer in writing
1–2	Axler <i>LADR</i> §5–7, §10.A; then N&C §2.1.8	Prove $\text{Tr}(T^N) = \lambda_+^N + \lambda_-^N$	Why does $e^{-\beta\hat{H}}$ need the spectral theorem to exist?
3	—	Run the notebook ; exercises 1–3	At what N does brute force stop on your laptop? Extrapolate to $N = 60$.
4	Shankar §21.2	Re-derive the Ising Z his way	What exactly is being Wick-rotated, and why does dimension go up by one?
5–6	Feynman ch. 1 (<i>The Partition Function</i>), ch. 2, ch. 5	Do §5.2 by hand; read §5.4 (Onsager) without trying to follow every step	Why is 1D solvable and 2D a landmark?
7–8	PBoC2 ch. 6 <i>Entropy Rules!</i>	Work the gene-expression model	Which of Part IX Fig. 2's six quantities can you now compute for a real cell?
9	N&C §2.4, §11.3, ch. 8	Write $\rho = e^{-\beta\hat{H}}/Z$ and verify $\text{Tr } \rho = 1$	Where does positivity get used, exactly?
10	Axler <i>MIRA</i> ch. 1–3 (or Le Gall)	Verify $\mu(dx) = e^{-\beta U} dx/Z$ is a probability measure	In what sense does a positive operator <i>determine</i> a measure?
11–12	Levin & Peres ch. 1–4 (the one book to acquire)	Notebook exercise 4: autocorrelation time vs ξ	Restate §6 in the language of mixing times.

Table 5. One book to acquire; eleven weeks from your own shelves. At the end you can read Part IX §7 on Boltzmann generators and know precisely what problem it solves.

IN PLAIN WORDS

The plan is one week longer than it needs to be on purpose. Weeks 4 and 5–6 cover the same calculation twice, from quantum mechanics and from statistical mechanics. That redundancy is the point: it is the difference between knowing a derivation and knowing an object.

11 Verification ledger

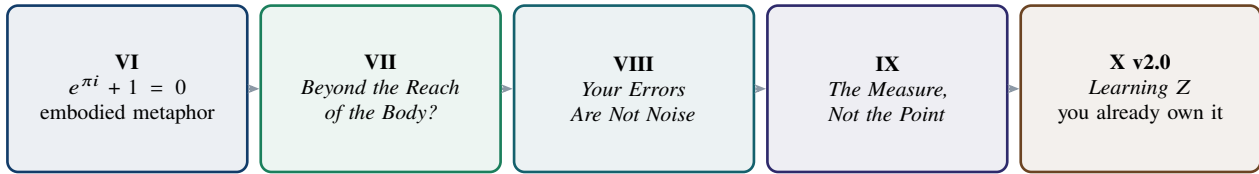
Every book below was opened and its page count read, on 19 September 2026.

Book	pp	Library / folder
Feynman, <i>Statistical Mechanics: A Set of Lectures</i>	371	Pure_Maths/Module_8
Le Gall, <i>Measure Theory, Probability and Stochastic Processes</i>	409	Pure_Maths/Module_8
Lauritzen, <i>Fundamentals of Mathematical Statistics</i>	—	Pure_Maths/Module_8
Axler, <i>Measure, Integration & Real Analysis</i> (2022)	430	Pure_Maths/Module_5
Rudin, <i>Principles of Mathematical Analysis</i>	—	Pure_Maths/Module_6
Axler, <i>Linear Algebra Done Right</i> (2015)	352	Pure_Maths/Module_1/Tensor-College
Kuttler, <i>Linear Algebra and Analysis</i> (2020)	—	Pure_Maths/Module_1/more-libgen
Shankar, <i>Principles of Quantum Mechanics</i> 2ed	696	TensoQi101/Main_Ref_Texts
Nielsen & Chuang, <i>QC and QI</i> (10th anniv.)	710	TensoQi101/Main_Ref_Texts
Schumacher & Westmoreland, <i>Quantum Processes...</i>	483	TensoQi101/Main_Ref_Texts
Sakurai & Napolitano, <i>Modern QM</i> 3ed (+2 soln.)	—	TensoQi101/Aux_Ref_Texts
4 DFT texts (incl. Harvard practical guide)	11–53	TensoQi101/PoQM/DFT_Density_Functional_Theory
Phillips et al., <i>Physical Biology of the Cell</i> 2ed	1089	QBIT-HIS/PBoC2
Bressloff, <i>Stochastic Processes in Cell Biology</i> Vol. 1	773	QBIT-HIS/PBoC2
Bressloff, Vol. 2	—	QBIT-HIS/PBoC2
Attard, <i>Quantum Statistical Mechanics from First Principles</i>	285	QBIT-HIS
Bru & de Siqueira Pedra, <i>C*-Algebras and Math. Found. of QSM</i>	497	QBIT-HIS
Khrennikov, <i>Open Quantum Systems in Biology...</i>	371	QBIT-HIS CONTESTED
Barato & Seifert, thermodynamic uncertainty relation	11	QBIT-HIS/PBoC2
Marais et al., <i>The future of quantum biology</i>	14	QBIT-HIS

11.1 Method

Claim	How it was checked
681 books across four libraries	find for *.pdf *.epub *.mobi *.djvu in each tree
“Shankar, 14 hits for <i>partition function</i> ”	pdftotext piped to grep -ic over every PDF >1 MB in TensoQi101
Shankar §21.2 quotations	extracted from the text layer; equation numbers 21.2.44 and 21.2.59 as printed
Feynman chapter and section titles	table of contents, text layer
Axler section titles and page numbers	table of contents, text layer
Tables 1–3, Figures 4–5 (§5–§6)	computed by the companion notebook; numpy 1.26.4, fixed seeds

11.2 The dossier series



The library was never the constraint.

681 books, four folders, and everything the partition function needs except one free PDF on mixing times. What was missing was not the books — it was knowing which three sections of which one to read first, and that 2^N configurations collapse to two numbers.